

# Assignment 1, Harmonic Analysis

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## Problem 1

(a)

*Proof.*

$$\begin{aligned} M'_c(f)(x) &= \sup_{r>0} \frac{1}{r^d} \int_{x+(-r/2, r/2)^d} |f(y)| dy \\ &\leq \sup_{r>0} \frac{1}{r^d} \int_{B(x, r\sqrt{d}/2)} |f(y)| dy && (x + (-r/2, r/2)^d \subseteq B(x, r\sqrt{d}/2)) \\ &= \sup_{\rho>0} \frac{d^{d/2}}{2^d \rho^d} \int_{B(x, \rho)} |f(y)| dy && (\rho = r\sqrt{d}/2) \\ &= d^{d/2} \cdot v_d \cdot 2^{-d} \sup_{\rho>0} \frac{1}{v_d \rho^d} \int_{B(x, \rho)} |f(y)| dy \\ &= d^{d/2} \cdot v_d \cdot 2^{-d} \cdot M_c(f)(x). \quad \square \end{aligned}$$

(b)

*Proof.*

$$\begin{aligned} M_c(f)(x) &= \sup_{r>0} \frac{1}{v_d r^d} \int_{B(x, r)} |f(y)| dy \\ &\leq \sup_{r>0} \frac{1}{v_d r^d} \int_{x+(-r, r)^d} |f(y)| dy && (B(x, r) \subseteq x + (-r, r)^d) \\ &= \sup_{s>0} \frac{2^d}{v_d s^d} \int_{x+(-s/2, s/2)^d} |f(y)| dy && (s = 2r) \\ &= 2^d \cdot v_d^{-1} \sup_{s>0} \frac{1}{s^d} \int_{x+(-s/2, s/2)^d} |f(y)| dy \\ &= 2^d \cdot v_d^{-1} \cdot M'_c(f)(x). \quad \square \end{aligned}$$

## Problem 2

(a)

*Proof.* Since clearly  $g_q(x) := |f(x) - q| \in L^1_{\text{loc}}(\mathbb{R}^d)$ , by Lebesgue Differentiation Theorem

$$\lim_{r \rightarrow 0^+} \frac{1}{|B(x, r)|} \int_{B(x, r)} |f(y) - q| dy = |f(x) - q| \quad \text{for a.e. } x \in \mathbb{R}^d. \quad (1)$$

Let  $A_q$  be the set where it fails, then  $|A_q| = 0$  and the limit holds for all  $x \in \mathbb{R}^d \setminus A_q$ .  $\square$

(b)

*Proof.* Let  $A := \bigcup_{q \in \mathbb{Q}} A_q$ , where each  $A_q$  is as in previous part. Since  $\mathbb{Q}$  is countable,  $|A| = 0$ , and for every  $x \in \mathbb{R}^d \setminus A$ , (1) holds simultaneously for all  $q \in \mathbb{Q}$ . Fix  $x \in \mathbb{R}^d \setminus A$  and  $\varepsilon > 0$ . Since  $f$  is real-valued and  $\mathbb{Q}$  is dense in  $\mathbb{R}$ , choose  $q \in \mathbb{Q}$  such that  $|f(x) - q| < \varepsilon/2$ .

$$\begin{aligned} & \limsup_{r \rightarrow 0^+} \frac{1}{|B(x, r)|} \int_{B(x, r)} |f(y) - f(x)| dy \\ & \leq \limsup_{r \rightarrow 0^+} \frac{1}{|B(x, r)|} \int_{B(x, r)} (|f(y) - q| + |f(x) - q|) dy \\ & = \limsup_{r \rightarrow 0^+} \frac{1}{|B(x, r)|} \int_{B(x, r)} |f(y) - q| dy + |f(x) - q| \quad (|f(x) - q| \text{ const. in } y) \\ & = |f(x) - q| + |f(x) - q| = 2|f(x) - q| < \varepsilon \end{aligned}$$

Since  $\varepsilon > 0$  was arbitrary,

$$\lim_{r \rightarrow 0^+} \frac{1}{|B(x, r)|} \int_{B(x, r)} |f(y) - f(x)| dy = 0 \quad \forall x \in \mathbb{R}^d \setminus A. \quad (2) \quad \square$$

(c)

*Proof.* Let  $A$  be the null set from part (b) and let  $r_n := \sqrt{d} \cdot 2^{-n}$ , so  $Q_n(x) \subseteq B(x, r_n)$ .

Fix  $x \in \mathbb{R}^d \setminus A$  and  $\varepsilon > 0$ , by (2),  $\exists N \in \mathbb{N}$  such that  $\forall n \geq N$ ,

$$\frac{1}{|B(x, r_n)|} \int_{B(x, r_n)} |f(y) - f(x)| dy < \frac{\varepsilon}{v_d d^{d/2}}.$$

Then for all  $n \geq N$ ,

$$\begin{aligned}
\left| \frac{1}{|Q_n|} \int_{Q_n} f(y) dy - f(x) \right| &= \left| \frac{1}{|Q_n|} \int_{Q_n} (f(y) - f(x)) dy \right| \\
&\leq \frac{1}{|Q_n|} \int_{Q_n} |f(y) - f(x)| dy \\
&\leq \frac{1}{|Q_n|} \int_{B(x, r_n)} |f(y) - f(x)| dy \\
&= \frac{|B(x, r_n)|}{|Q_n|} \cdot \frac{1}{|B(x, r_n)|} \int_{B(x, r_n)} |f(y) - f(x)| dy \\
&= \frac{v_d r_n^d}{2^{-nd}} \cdot \frac{1}{|B(x, r_n)|} \int_{B(x, r_n)} |f(y) - f(x)| dy \\
&= \frac{v_d \cdot d^{d/2} \cdot 2^{-nd}}{2^{-nd}} \cdot \frac{1}{|B(x, r_n)|} \int_{B(x, r_n)} |f(y) - f(x)| dy \\
&= v_d d^{d/2} \cdot \frac{1}{|B(x, r_n)|} \int_{B(x, r_n)} |f(y) - f(x)| dy \\
&< v_d d^{d/2} \cdot \frac{\varepsilon}{v_d d^{d/2}} = \varepsilon.
\end{aligned}$$

Since  $\varepsilon > 0$  was arbitrary,  $\left| \frac{1}{|Q_n|} \int_{Q_n} f(y) dy - f(x) \right| \rightarrow 0$  as  $n \rightarrow \infty$ , which is equivalent to  $\frac{1}{|Q_n(x)|} \int_{Q_n(x)} f(y) dy \rightarrow f(x)$  for all  $x \in \mathbb{R}^d \setminus A$ .  $\square$

### Problem 3